

Question

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4. CD

Op-code	Operand	Description
0xB	RXY	JUMP to the instruction located in the memory cell at address XY if the bit pattern in register R is equal to the bit pattern in register number 0. Otherwise, continue with the normal sequence of execution. (The jump is implemented by copying XY into the program counter during the execute phase.) <i>Example:</i> 0xB43C would first compare the contents of register 0x4 with the contents of register 0x0. If the two were equal, the pattern 0x3C would be placed in the program counter so that the next instruction executed would be the one located at that memory address. Otherwise, nothing would be done and program execution would continue in its normal sequence.

Question

7. The following are instructions written in Vole machine language. Translate them into English.
- | | | |
|-----------|-----------|-----------|
| a. 0x7123 | b. 0x40E1 | c. 0xA304 |
| d. 0xB100 | e. 0x2BCD | |

Question

7. The following are instructions written in Vole machine language. Translate them into English.

- | | | |
|-----------|-----------|-----------|
| a. 0x7123 | b. 0x40E1 | c. 0xA304 |
| d. 0xB100 | e. 0x2BCD | |

7. a. OR the contents of register 2 with the contents of register 3 and place the result in register 1.
- b. Move the contents of register E to register 1.
- c. Rotate the contents of register 3 four bits to the right.
- d. Compare the contents of registers 1 and 0. If the patterns are equal, jump to the instruction at address 00. Otherwise, continue with the next sequential instruction.
- e. Load register B with the value (hexadecimal) CD.

Question

9. Translate the following instructions from English into Vole.
- a. **LOAD** register 0x6 with the value 0x77.
 - b. **LOAD** register 0x7 with the contents of memory cell 0x77.
 - c. **JUMP** to the instruction at memory location 0x24 if the contents of register 0x0 equals the value in register 0xA.
 - d. **ROTATE** register 0x4 three bits to the right.
 - e. **AND** the contents of registers 0xE and 0x2 leaving the result in register 0x1.

Question

9. Translate the following instructions from English into Vole.
- a. **LOAD** register 0x6 with the value 0x77.
 - b. **LOAD** register 0x7 with the contents of memory cell 0x77.
 - c. **JUMP** to the instruction at memory location 0x24 if the contents of register 0x0 equals the value in register 0xA.
 - d. **ROTATE** register 0x4 three bits to the right.
 - e. **AND** the contents of registers 0xE and 0x2 leaving the result in register 0x1.

9. a. 2677 b. 1677 c. BA24 d. A403 e. 81E2

Question

- **12.** Suppose the memory cells at addresses 0x00 through 0x03 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x26
0x01	0x55
0x02	0xC0
0x03	0x00

- Translate the first instruction into English.
- If the machine is started with its program counter containing 0x00, what bit pattern is in register 0x6 when the machine halts?

Question

12. Suppose the memory cells at addresses 0x00 through 0x03 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x26
0x01	0x55
0x02	0xC0
0x03	0x00

- Translate the first instruction into English.
- If the machine is started with its program counter containing 0x00, what bit pattern is in register 0x6 when the machine halts?

12. a. Place the value 55 in register 6. b. 55

Question

14. Suppose the memory cells at addresses 0x00 through 0x05 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x12
0x01	0x02
0x02	0x32
0x03	0x42
0x04	0xC0
0x05	0x00

When answering the following questions, assume that the machine starts with its program counter equal to 0x00.

- Translate the instructions that are executed into English.
- What bit pattern is in the memory cell at address 0x42 when the machine halts?
- What bit pattern is in the program counter when the machine halts?

Question

14. Suppose the memory cells at addresses 0x00 through 0x05 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x12
0x01	0x02
0x02	0x32
0x03	0x42
0x04	0xC0
0x05	0x00

When answering the following questions, assume that the machine starts with its program counter equal to 0x00.

- Translate the instructions that are executed into English.
 - What bit pattern is in the memory cell at address 0x42 when the machine halts?
 - What bit pattern is in the program counter when the machine halts?
14. a. Load register 2 with the contents of memory cell 02.
Store the contents of register 2 in memory cell 42.
Halt.
- 32
 - 06

Question

15. Suppose the memory cells at addresses 0x00 through 0x09 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x1C
0x01	0x03
0x02	0x2B
0x03	0x03
0x04	0x5A
0x05	0xBC
0x06	0x3A
0x07	0x00
0x08	0xC0
0x09	0x00

Assume that the machine starts with its program counter containing 0x00.

- What will be in the memory cell at address 0x00 when the machine halts?
- What bit pattern will be in the program counter when the machine halts?

Question

15. Suppose the memory cells at addresses 0x00 through 0x09 in the Vole contain the following bit patterns:

Address	Contents
0x00	0x1C
0x01	0x03
0x02	0x2B
0x03	0x03
0x04	0x5A
0x05	0xBC
0x06	0x3A
0x07	0x00
0x08	0xC0
0x09	0x00

Assume that the machine starts with its program counter containing 0x00.

- What will be in the memory cell at address 0x00 when the machine halts?
- What bit pattern will be in the program counter when the machine halts?

Question

18. Suppose the memory cells at addresses 0xF0 through 0xFD in the Vole contain the following (hexadecimal) bit patterns:

Address	Contents
0xF0	0x20
0xF1	0x00
0xF2	0x22
0xF3	0x02
0xF4	0x23
0xF5	0x04
0xF6	0xB3
0xF7	0xFC
0xF8	0x50
0xF9	0x02
0xFA	0xB0
0xFB	0xF6
0xFC	0xC0
0xFD	0x00

If we start the machine with its program counter containing 0xF0, what is the value in register 0x0 when the machine finally executes the halt instruction at location 0xFC?

Question

18. Suppose the memory cells at addresses 0xF0 through 0xFD in the Vole contain the following (hexadecimal) bit patterns:

Address	Contents
0xF0	0x20
0xF1	0x00
0xF2	0x22
0xF3	0x02
0xF4	0x23
0xF5	0x04
0xF6	0xB3
0xF7	0xFC
0xF8	0x50
0xF9	0x02
0xFA	0xB0
0xFB	0xF6
0xFC	0xC0
0xFD	0x00

If we start the machine with its program counter containing 0xF0, what is the value in register 0x0 when the machine finally executes the halt instruction at location 0xFC?

18. 04. The program is a loop that is terminated when the value in register 0 (initiated at 00) is finally incremented by twos to the value in register 3 (initiated at 04).

Question

20. Suppose the memory cells at addresses 0x20 through 0x28 in the Vole contain the following bit patterns:

Address	Contents
0x20	0x12
0x21	0x20
0x22	0x32
0x23	0x30
0x24	0xB0
0x25	0x21
0x26	0x24
0x27	0xC0
0x28	0x00

Assume that the machine starts with its program counter containing 0x20.

- What bit patterns will be in registers 0x0, 0x1, and 0x2 when the machine halts?
- What bit pattern will be in the memory cell at address 0x30 when the machine halts?
- What bit pattern will be in the memory cell at address 0xB0 when the machine halts?

Question

20. Suppose the memory cells at addresses 0x20 through 0x28 in the Vole contain the following bit patterns:

Address	Contents
0x20	0x12
0x21	0x20
0x22	0x32
0x23	0x30
0x24	0xB0
0x25	0x21
0x26	0x24
0x27	0xC0
0x28	0x00

Assume that the machine starts with its program counter containing 0x20.

- What bit patterns will be in registers 0x0, 0x1, and 0x2 when the machine halts?
- What bit pattern will be in the memory cell at address 0x30 when the machine halts?
- What bit pattern will be in the memory cell at address 0xB0 when the machine halts?

20. The point to this problem is that a bit pattern stored in memory is subject to interpretation—it may represent part of the operand of one instruction and the op-code field of another.

- Registers 0, 1, and 2 will contain 32, 24, and 12, respectively.
- 12
- 32

Question

- 23.** In each of the following cases, write a short program in Vole to perform the requested activities. Assume that each of your programs is placed in memory starting at address 0x00.
- Move the value at memory location 0xD8 to memory location 0xB3.
 - Interchange the values stored at memory locations 0xD8 and 0xB3.
 - If the value stored in memory location 0x44 is 0x00, then place the value 0x01 in memory location 0x46; otherwise, put the value 0xFF in memory location 0x46.

Question

23. In each of the following cases, write a short program in Vole to perform the requested activities. Assume that each of your programs is placed in memory starting at address 0x00.
- a. Move the value at memory location 0xD8 to memory location 0xB3.
 - b. Interchange the values stored at memory locations 0xD8 and 0xB3.
 - c. If the value stored in memory location 0x44 is 0x00, then place the value 0x01 in memory location 0x46; otherwise, put the value 0xFF in memory location 0x46.

23.	a.	b.	c.
	14D8	14D8	2000
	34B3	15B3	1144
	C000	358D	B10A
		34BD	22FF
		C000	B00C
			2201
			3246
			C000

Question

- *32.** Suppose the registers 0x4 and 0x5 in the Vole contain the bit patterns 0x3A and 0xC8, respectively. What bit pattern is left in register 0x0 after executing each of the following instructions:
- a. 0x5045 b. 0x6045 c. 0x7045
 - d. 0x8045 e. 0x9045

Question

- *32.** Suppose the registers 0x4 and 0x5 in the Vole contain the bit patterns 0x3A and 0xC8, respectively. What bit pattern is left in register 0x0 after executing each of the following instructions:
- a. 0x5045 b. 0x6045 c. 0x7045
 - d. 0x8045 e. 0x9045

32. a. 02 b. AC c. FA d. 08 e. F2